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CS 461 Senior Capstone

RFC Draft

1. Problem and Context

Therapist Flow is a burnout prevention scheduler for therapists and their supervisors. Before the first build sprint, we must decide whether it ships as a mobile application or a website. The product asks therapists for a short daily mood check, records their sessions, and shows how workload affects burnout risk.

The decision. Whether Therapist Flow is delivered as an installable mobile application or a website. We also evaluate a terminal-based tool and doing nothing.

Why now. The story map's walking skeleton is set: daily mood survey, session logging, workload overview, and a rebalancing plan. Each of those behaves differently on a phone than in a browser, so the team cannot estimate, design, or test until the platform is fixed. The partner clinic also expects a supervisor pilot next term.

Do-nothing consequence. Without a decision we cannot choose a framework, set up CI, or write acceptance criteria. Designers would mock up screens for unknown devices. Meanwhile therapists keep judging workload from memory and spreadsheets, the habit this product exists to change.

Reversibility. Hard to reverse once started. The platform determines the codebase, reminder mechanism, local storage, authentication, and release process. Switching after launch means rewriting the client, although the backend API and data model would survive.

Inherited constraints. None: no legacy client, hosting commitment, or platform requirement from the partner. Two stories in the map are still unconfirmed assumptions (marked "?"): blocking sessions when workload is too high, and the action plan. Neither changes the platform choice, but both are open questions for the partner.

2. Options

Option A: Mobile application. A cross-platform app for iOS and Android with local storage and push notifications. The daily mood survey may arrive as a reminder, and a therapist can log a session between appointments in under a minute. Supervisors use the same app with an aggregated team view.

Option B: Website. A responsive web app served from standard cloud hosting. Therapists open it in a browser or add it to a home screen. Reminders rely on email or browser notifications, and updates ship instantly with no store review.

Option C: Terminal-based tool. A command-line client distributed as a single binary. It would be quick to build and easy to test, and could call the same backend API. Workload charts would be rendered as text.

Option D: Do nothing. Defer the choice, continue with backend and data-model work only, and let therapists keep using calendars and spreadsheets.

3. Analysis

The tradeoffs that matter are reminder reliability, privacy, offline use, complexity, and cost.

Mobile application. Daily mood tracking works only if the prompt reaches the therapist reliably. Apple's and Google's documentation describe push notifications and local scheduled notifications as first-class features of installed apps, and local notifications fire without a network connection. In our spike, 30 simulated therapists received the 5 p.m. mood reminder on 98 percent of days over two weeks. Local encrypted storage also lets a therapist log a session in a basement office with no signal and sync later. The cost is complexity: app-store review, OS version fragmentation, and a slower release cadence.

Website. The web has the cheapest distribution and fastest releases. But browser push is unavailable on iOS unless the site is installed to the home screen, and Safari's documented storage policy allows it to clear site data after about seven days without use. In our spike, only 71 percent of reminders were delivered on iPhones, and 9 of 40 offline session logs were lost after a device slept over a weekend. A tool for preventing burnout cannot depend on a reminder that a quarter of users never see.

Terminal-based tool. It is the cheapest to build, but it does not fit the user. In a short poll of 18 clinicians at the partner clinic, 17 said they would check in on a phone, and none had a terminal installed. It cannot send reminders by itself. We keep it only as a possible internal admin tool.

Do nothing. Costs nothing now. But in the same poll, 12 of 18 clinicians said they could not state their weekly session count without checking a calendar, and 7 had worked more than 30 client hours in a week at least once this term.

Privacy and security. Mood entries are sensitive. A native app can use the device keystore and biometric unlock, whereas a website relies on session cookies and browser storage. Both can meet our requirements, but the app makes it easier to keep data encrypted on the device.

4. Recommendation and Verification Plan

Recommendation. Build Therapist Flow as a cross-platform mobile application (Option A) using React. Build the reminder, offline logging, and sync layer first, because the walking skeleton depends on them. A web dashboard for supervisors is deferred to a later release.

Verification. The team will know the choice was right if the pilot with 20 therapists and 4 supervisors meets these thresholds:

- **Reminder delivery:** at least 95 percent of daily mood reminders delivered, measured from notification logs over four weeks.
- **Mood completion:** at least 70 percent of therapists complete the daily survey on 5 of 7 days by week four.
- **Logging speed:** median session logging time at or below 45 seconds.
- **Sync reliability:** 99.9 percent of offline entries synced without loss or duplication, across 1,000 automated test entries.
- **Stability:** at least 99.5 percent crash-free sessions.
- **Usefulness:** at least 60 percent of therapists open the workload overview weekly, and at least half of supervisors act on a risk alert within a week.
- **Delivery:** a pilot-ready build within 12 weeks of kickoff.

Failure signals. Any of these means the plan is not working: reminder delivery below 85 percent after two fix cycles; mood completion below 40 percent at week four; sync loss above 0.5 percent; app-store rejection delaying two consecutive releases by more than a week; or therapists reporting that they dislike installing another app. That last signal matters: if installs stall below 50 percent of invited users, we treat it as evidence for Option B.

Reversal trigger and cost. We revisit the decision at the week-six checkpoint if two or more failure signals appear. Reversal means building a web client against the same API, estimated at 12 to 16 engineer-weeks, and discarding about 40 percent of the client code. The backend, data model, risk-scoring logic, and designs carry over. The cheaper fallback is wrapping a web app for the stores (Capacitor), at about six engineer-weeks.

5. Constraints

Licensing. React Native, SQLite, and the charting and sync libraries are MIT or Apache 2.0 licensed, so they allow commercial use without copyleft obligations. CI will run a license scan and block GPL or AGPL packages. Therapist Flow itself will be proprietary under our standard license.

Data use and privacy. The design follows Privacy by Design: collect only what is needed, default to the most private setting, and build protection into the architecture. Mood entries and session counts belong to the therapist. Supervisors see only team-level aggregates unless a therapist opts in to share individual data. Sessions are logged with an anonymous case code, never client names or clinical notes, which keeps the product outside most health-record

requirements. We will confirm that with the partner's compliance officer. Local data is encrypted at rest, and it is deleted from a device on sign-out.

IP ownership. Code written by team members is owned by the company under existing agreements. Burnout-risk scoring will use published, open instruments as inputs (for example, workload and exhaustion measures from the research literature); we will cite them rather than copy proprietary questionnaires.

Partner confidentiality. The partner clinic's staffing data and poll responses are under NDA. This RFC uses aggregated figures only. Clinic name, staff names, schedules, and caseload details have been removed, and no partner data is stored in the repository.

6. Delegation and Validation

What I delegated. I used AI tools for four tasks: summarizing platform documentation on notifications and offline storage; drafting the first version of this RFC and its comparison table; generating a reminder-delivery test harness for the spike; and producing a first-pass dependency license summary.

What made that reasonable. Code went through pull-request review, automated tests, and CI, and the spike ran on a staging environment with synthetic users and no real clinical data. Rollback was trivial because the spike lived on a branch with no production dependencies. Research and prose carried more risk, so I checked them against primary sources before using them.

Where I stopped and checked. I re-ran the reminder and offline-sync tests myself on two physical phones, one iOS and one Android, instead of trusting the generated results. I read Apple's and Google's notification and storage documentation directly to confirm the claims about iOS web push and data eviction. I checked each dependency's license in its own repository, and recalculated the engineer-week estimates with the team lead.

What I changed or rejected. The first draft recommended a website because of lower cost, but it relied on general claims about progressive web apps that did not match our iPhone results, so I rejected that conclusion. I corrected two test cases that did not simulate a sleeping device realistically, removed a suggested library with an unclear license, and cut unsupported performance figures. I also removed a suggestion to show supervisors individual mood scores by default, which conflicted with our privacy principle.